

Mistral-7b Complete Guide on Colab



Yogendra Sisodia · [Follow](#)

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 MISTRAL AI_ **7B Complete Guide** on **colab**



LangChain

Lang Chain Integration



Chroma

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The performance of Mistral 7B surpasses that of Llama 2 13B across all criteria and is comparable to Llama 34B. Furthermore, it exhibits

significant superiority in terms of code quality and logical analysis benchmarks.

An LLM that is open source provides a level of transparency with respect to its operational mechanisms, architectural design, training data, and methodology, as well as its utilization. Moreover, the efficient optimization of an open-source LLM like Mistral-7B has the potential to decrease latency and enhance performance.

I am excited to explore Mistral-7B instruct, and I am covering the following technologies in this article, along with a walkthrough video:

1. The Lang-Chain framework and integration with Mistral-7B-Instruct.
2. Retrieval Augmented Generation (RAG) with Mistral-7B-Instruct and Chroma DB.
3. GGUF format for commodity hardware (Running on CPU).

Colab Snippets

```
### Installation on Colab (As Of Oct)
!pip install -q -U bitsandbytes
!pip install -q -U git+https://github.com/huggingface/transformers.git
!pip install -q -U git+https://github.com/huggingface/peft.git
!pip install -q -U git+https://github.com/huggingface/accelerate.git
!pip install -q -U einops
!pip install -q -U safetensors
!pip install -q -U torch
!pip install -q -U xformers
!pip install -q -U langchain
!pip install -q -U ctransformers[cuda]
```

```
!pip install chromadb
!pip install sentence-transformers
```

Lang-Chain Integration and BitsAndBytes

```
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```

```
import torch
from transformers import BitsAndBytesConfig
from langchain import HuggingFacePipeline
from langchain import PromptTemplate, LLMChain
llm = HuggingFacePipeline(pipeline=pipeline)

quantization_config = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_compute_dtype=torch.float16,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_use_double_quant=True,
)
model_id = "mistralai/Mistral-7B-Instruct-v0.1"

from transformers import AutoModelForCausalLM, AutoTokenizer, pipeline
model_4bit = AutoModelForCausalLM.from_pretrained(model_id, device_map="auto")
tokenizer = AutoTokenizer.from_pretrained(model_id)

pipeline = pipeline(
    "text-generation",
    model=model_4bit,
    tokenizer=tokenizer,
    use_cache=True,
    device_map="auto",
    max_length=500,
    do_sample=True,
    top_k=5,
    num_return_sequences=1,
    eos_token_id=tokenizer.eos_token_id,
    pad_token_id=tokenizer.eos_token_id,
)

#### Prompt
template = """<s>[INST] You are a helpful, respectful and honest assistant.
Answer the question below from context below :
{context}
{question} [/INST] </s>
"""
```

```

question_p = """What is the date for announcement"""
context_p = """ On August 10 said that its arm JSW Neo Energy has agreed to l
prompt = PromptTemplate(template=template, input_variables=["question","cont
llm_chain = LLMChain(prompt=prompt, llm=llm)
response = llm_chain.run({"question":question_p,"context":context_p})
response

```

Retrieval Augmented Generation (RAG) with Chroma-DB

```

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```

```

import chromadb
from chromadb.config import Settings
from langchain.llms import HuggingFacePipeline
from langchain.document_loaders import TextLoader
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain.embeddings import HuggingFaceEmbeddings
from langchain.chains import RetrievalQA
from langchain.vectorstores import Chroma

```

```

mna_news = """JSW Energy on August 10 said that its arm JSW Neo Energy has a
Sajjan Jindal led JSW group had signed an exclusivity agreement with Hyderab
This is the largest acquisition by JSW Energy since inception, which compris
With this acquisition, JSW Energy's current operational generation capacity
"""

```

```

from langchain.schema.document import Document
documents = [Document(page_content=mna_news, metadata={"source": "local"})]
#####
text_splitter = RecursiveCharacterTextSplitter(chunk_size=500, chunk_overlap
all_splits = text_splitter.split_documents(documents)
model_name = "sentence-transformers/all-mpnet-base-v2"
model_kwargs = {"device": "cuda"}
embeddings = HuggingFaceEmbeddings(model_name=model_name, model_kwargs=model
#####
vectordb = Chroma.from_documents(documents=all_splits, embedding=embeddings,
#####
retriever = vectordb.as_retriever()
#####
qa = RetrievalQA.from_chain_type(
    llm=llm,
    chain_type="stuff",
    retriever=retriever,
    verbose=True

```

```

)

def run_my_rag(qa, query):
    print(f"Query: {query}\n")
    result = qa.run(query)
    print("\nResult: ", result)

### Ask Queries Now
query = """ What company is buyer and seller here """
run_my_rag(qa, query)

```

GGUF Format

```

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from langchain.llms import CTransformers
config = {'max_new_tokens': 100, 'temperature': 0}
llm = CTransformers(model='TheBloke/Mistral-7B-Instruct-v0.1-GGUF',model_file=

template = """<s>[INST] You are a helpful, respectful and honest assistant.
Answer the question below from context below :
{context}
{question} [/INST] </s>
"""

#### Prompt
question_p = """What is the date for announcement"""
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prompt = PromptTemplate(template=template, input_variables=["question","cont
llm_chain = LLMChain(prompt=prompt, llm=llm)
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response

```

Conclusion

Organizations have the ability to incorporate additional functionalities into the open-source LLM that might cater to their particular requirements. Furthermore, the LLMs can be trained using specific

datasets. Implementing modifications or requirements on a proprietary LLM system necessitates collaboration with a vendor, resulting in the expenditure of both time and financial resources. Mistral-7B is definitely a strong candidate for the open-source community and I am excited.

Resources (Colab and Walkthrough Video)

Google Colaboratory Edit description colab.research.google.com	
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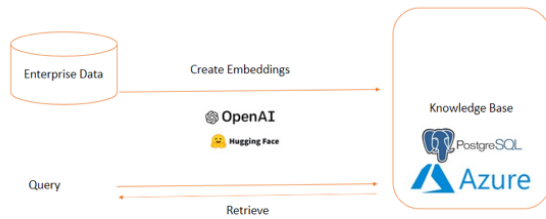
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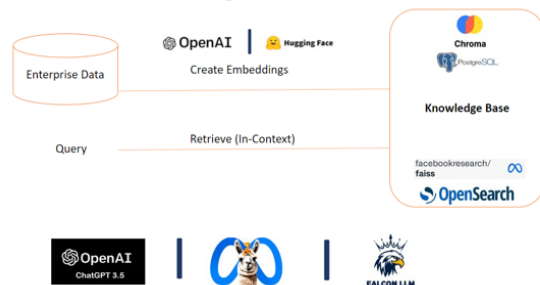
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